

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #22 – Smart Parking Space Rental & License Validator

Name: Dhanya S Prasad

SRN: PES1UG24CS149

Section: C

Use-Case Flow Specification

Use Case ID: UC-05	Use Case Name: Process Payment
Primary Actor: Vehicle Owner	Secondary Actors: Payment Gateway, Gate Camera System, Parking Warden

Preconditions

1. The vehicle owner has an active reservation or a valid parked session recorded against a scanned license plate.
2. The vehicle's entry time was already logged by the Gate Camera System at the entry gate when the vehicle first arrived (Use Case UC-02: Scan Vehicle License Plate). This is a separate scan event from the exit-gate scan described in this flow.
3. The Payment Gateway service is reachable and operational.

Postconditions

1. The computed parking fee has been charged successfully and recorded against the vehicle's session.
2. A digital receipt has been generated and sent to the vehicle owner.
3. The exit gate is authorized to open and the parking bay is marked as available.

Main Success Scenario

1. The vehicle owner drives to the exit gate; the Gate Camera System scans the license plate.
2. The system retrieves the logged entry time and validates the plate against the reservation records (includes UC-03: Validate License Plate).
3. The system calculates the elapsed parked duration and computes the total fee using the tiered hourly rate (includes UC-04: Calculate Parking Fee).
4. The system displays the computed fee to the vehicle owner on the kiosk screen or mobile app.
5. The vehicle owner selects a payment method: credit/debit card or UPI/mobile wallet.
6. The system forwards the payment request to the Payment Gateway.
7. The Payment Gateway authorizes the transaction and returns a success response.
8. The system generates a digital receipt and emails or displays it to the vehicle owner.
9. The exit gate opens automatically and the parking bay status is updated to “available”.
10. The use case ends successfully.

Alternate Flow – 7a. Payment Declined

- 7a.1. The Payment Gateway declines the transaction and returns a failure response.
- 7a.2. The system displays a “Payment Failed” message and prompts the vehicle owner to retry or choose another payment method.
- 7a.3. The vehicle owner selects a different payment method and the flow resumes from Main Success Scenario step 6.
- 7a.4. If payment fails again after two total attempts, the system cancels the automatic exit process, notifies the vehicle owner to proceed to the manual payment counter, and alerts the Parking Warden.